Highlights

cert-machine: independent three-valued certification of machine-proposed numerical claims in exact and interval arithmetic[⋆]

Carlos Toledo, Pedro Veras Guimarães

- Machine-proposed numbers are decided by a checker that shares no code with the claimant.
- Three verdicts, each published with the object that justifies it; a refusal names its reason.
- Soundness proved under the inclusion property; every function enclosure carries a stated tail bound.
- A pre-registered benchmark of 572 claims of known truth: no false verdict; two defects found and fixed.
- Exact certificates re-check without the engine; interval ones re-decide from pinned inputs.

cert-machine: independent three-valued certification of machine-proposed numerical claims in exact and interval arithmetic

Carlos Toledo[a,*], Pedro Veras Guimarães[b]

[a]*Independent researcher, the cert-machine project, Brazil*
[b]*Department of Mechanical Engineering, Universidade Federal de Santa Catarina, Florianópolis, Brazil*

Abstract

Search programs, surrogate models and large language models now propose constants, counterexamples, fitted parameters and operating points faster than they can be refereed, and each usually arrives with a checker written by its producer. We present cert-machine, a system that decides such claims independently of the claimant. A claim is read as a conjunction of inequalities over a declared box of inputs, under a universal or an epistemic reading, and receives one of three verdicts—certified, refuted or refused—each published with the object that justifies it: a certificate with its falsifier, a falsifying input, or, for a refusal, its reason, with, for a split refusal, two proved witnesses and, where a parameter is declared, the threshold at which the verdict would change. Verdicts are proved sound under the inclusion property of interval extensions, which the system secures with exact rational arithmetic and outward-rounded IEEE-754 interval arithmetic; the elementary and special functions it uses carry explicit tail bounds, stated as lemmas. Classical certifiers (the Krawczyk operator, second-order tests for maxima, boundary proofs, multilinear reductions, filtered exact geometric predicates, rational sums of squares and Chebyshev–Sturm minimisation) reduce analytic questions to finitely many such decisions, and a verification

[*]Draft v0.8 of 2026-10-21, built from repository commit aae61990d737; it will be archived with the next release of the repository, under the concept doi:10.5281/zenodo.22225860 (the previous version is in release v2026.10.21, doi:10.5281/zenodo.23551746). The measured numbers and counts set as macros are read at build time from the records named in the Data availability statement, and a record that no longer supports the sentence quoting it stops the build. Three companion papers carry the use cases in full (the kissing configurations, the registry's certificates, the mathematics release); the engineering demonstrator and the metocean study are the companions on the engineering demonstrator [1] and the metocean study [2].

[*]Corresponding author.

Email addresses: carlos@carlostoledo.co (Carlos Toledo), pedro.veras.guimaraes@ufsc.br (Pedro Veras Guimarães)

protocol (pinned inputs, mandatory adversarial controls, second implementations and in-browser re-execution) guards the code on which soundness rests. We evaluate the system on a pre-registered benchmark of 572 claims of known truth, built by mutating 97 published objects of its public register with an exact independent witness for each: 0 false certifications and 0 false refutations, 6 refusals, and two defects of the system's own found by the benchmark's rule and fixed on the record; on 1,012,121 comparisons of its interval kernel with an independent library, with no enclosure that failed to contain the exact value; and on a public register of 132 machine-generated claims from 59 claimants, whose refutations are the verifier's kind of defect. The contribution is the system and its semantics, not interval arithmetic, which is standard.

Keywords: interval arithmetic, validated numerics, computer-assisted proof, certificates, decision procedures, machine-generated mathematics, reproducibility

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1. Introduction

Context. Machines now propose mathematics and engineering numbers at a rate no referee can follow. Conjectures on fundamental constants are generated from numerical coincidences [3]; constructions in combinatorics and analysis are found by program search guided by language models [4, 5, 6] and by learned search [7, 8, 9]; autonomous research systems publish record configurations with their own verification files [10]; long proofs arrive with machine formalisations [11, 12, 13]; and in engineering, surrogate models, optimisers and language models propose operating points and design values [1]. In every case the number reaches its reader together with a checker written by its producer.

Problem. A checker written by the producer shares the producer's blind spots. The published record already shows the cost: a floating-point product printed as an exact constant, witnesses that hold only within the verifier's own tolerance, a checker that compares in double precision what it encloses at a hundred digits, a pairwise condition proved by structure and never checked pair by pair, and a clause present in a paper's theorem but absent from the formal statement that was proved [14, 15]. What is missing is a decider that reads the claim from its published statement and bytes, never runs the claimant's program in its trust path, and returns an answer that a third party can re-check without trusting the decider either.

State of the art. The ingredients of such a decider are decades old. Interval arithmetic with outward rounding provides enclosures with the inclusion property [16, 17, 18], now standardised [19, 20] and tested for conformance [21, 22]; the Krawczyk operator proves existence and uniqueness

of zeros [23, 24]; verification methods in floating point [24, 25, 26], ball arithmetic [27, 28] and interval global optimisation [29, 30, 31, 32] made computer-assisted proofs routine [33, 34, 35, 36, 37, 38]; and filtered exact predicates decide every sign in computational geometry [39, 40, 41]. Closest to the present work in semantics are δ-complete decision procedures for nonlinear real arithmetic [42, 43, 44], which answer "unsatisfiable" or "δ-satisfiable" with a witness box and from whose traces checkable proofs can be generated [45], and interval constraint propagation solvers that return boxes of unknown status [46]. Proof assistants check nonlinear inequalities with Taylor-interval and sum-of-squares certificates inside a trusted kernel [47, 48, 49, 50], but they cannot certify that a formal statement is the one intended [15, 51]. For outputs of language-model pipelines, certificate architectures that treat abstention as a first-class outcome have recently been proposed [52], and test batteries sound by construction, whose reported defects are never spurious, have been built [53].

Gap. These tools answer a question posed by their own user about their own model. None of them, to our knowledge, is organised around deciding *other parties'* published numbers: reading a printed digit string as the box it denotes, distinguishing a claim about every input in a box from a claim about an unknown true input inside it, and turning a failure to decide into a published result that states what would decide it. In engineering practice this second, epistemic reading is the usual one, and neither verification and validation nor uncertainty quantification [54] answers it: the former qualifies a model, the latter gives a dispersion, whereas the engineer asks whether a given rule holds for every input that the declared evidence allows.

Objective. This paper asks three questions. (Q1) Can a single three-valued decision grammar, proved sound under one hypothesis, cover claims arriving from mathematics, from engineering and from metocean statistics? (Q2) What must the arithmetic and the software process guarantee for that hypothesis to hold in an implementation? (Q3) On claims whose truth is known, how often does the system certify, refute or refuse, and at what cost?

Contributions. The contribution is a system, not a numerical technique:

1. a three-valued decision grammar with witnesses, refusal as a result and flip thresholds, with a universal and an epistemic reading, and its soundness theorem (Section 2.2.2);
2. an arithmetic layer that secures the theorem's single hypothesis on IEEE-754 doubles and exact rationals, with a lemma for each elementary and special function it uses (Section 2.2.1, Appendix A);
3. a catalogue of standard certifiers, each mapped to the grammar with its witness and its refusal reasons (Section 2.2.4);

4. a verification protocol whose rules are executable gates rather than guidelines, with the gates that are not yet enforced listed as such (Section 2.4.2); and
5. a pre-registered evaluation of the system against claims of known truth, built by mutating its own public register with independent exact witnesses, which found two defects of the system by its rule and measures what soundness promises (Section 2.3, Section 3).

Interval arithmetic, the Krawczyk operator, filtered predicates, sums of squares and Sturm sequences are cited, not claimed.

Outline. Section 2 describes the method: its foundations in the literature, the mathematical formulation (arithmetic, verdicts, soundness and termination, certifiers), the comparison metrics and decision rules, and the software architecture with its verification protocol. Section 3 reports the benchmark with known truth, the kernel's comparison with an independent library and the public register with its use cases; Section 4 discusses what the results show, counts the independence the system can claim, places it beside existing practice and states the threats to validity and the limits; Section 5 concludes. The appendices hold the reproduction table, the arithmetic in detail with the function lemmas, the thresholds, the certifiers in detail and the decision loop.

2. Methodology

The methodology is presented in four layers, from the mathematics on which a verdict rests to the code that computes it. Section 2.1 reviews the results the system is built on. Section 2.2 states the mathematical formulation: the arithmetic, the claims and their verdicts, the soundness and termination results, and the certifiers. Section 2.3 defines the metrics by which verdicts are compared with ground truth and with other tools, and the decision rules, each with its mathematical basis. Only then does Section 2.4 describe the software architecture and the protocol that guards it.

Four implementations recur below as running instances. (a) The *flowline gate* of the engineering demonstrator [1] decides, under the epistemic reading, whether the wellhead pressure of a water-injection line stays below a rule over a declared envelope of roughness, viscosity, diameter, density, elevation and length, with Colebrook's friction law solved inside the enclosure; the pump pressure is its operating choice. (b) The *rock-physics engine* [55] decides, over declared boxes of rock and fluid properties, whether a time-lapse seismic change is guaranteed to be detectable. (c) The *likelihood certifier* of the metocean study [2] proves the maximum-likelihood fits of extreme-value families to a buoy's hourly record and ranks the families. (d) A *universal branch-and-bound* decides a statement for every input in a

box by bisection on the widest coordinate; its instance in the register bounds, for every monic polynomial of a given degree with all roots in $[-1, 1]$, the measure of the set where the polynomial is below one in absolute value, by subdividing the box of root configurations until a certified upper bound falls below the bar [14].

2.1. Foundations

The system rests on interval analysis. Since Moore's formalisation of arithmetic on intervals [16, 17], its central result—the *fundamental theorem of interval arithmetic*, by which the interval evaluation of an expression encloses the range of the function it denotes—has turned finite floating-point computation into proof [18, 17]. The arithmetic is standardised as IEEE 1788-2015 [19], test suites check the conformance of libraries to it [21], and directed rounding can be emulated when the rounding mode cannot be set [26], the situation of the implementation described here (Proposition 1); the interval layer is not claimed conformant, and no conformance suite has been run on it (Section 4.5). Verification methods carry the theorem into floating point with directed rounding [24, 25, 27], interval branch-and-bound turns enclosures into rigorous global optimisation [29, 30, 31, 32], and the same arithmetic underlies computer-assisted proofs from dynamics to the Kepler conjecture and partial differential equations [33, 56, 34, 37, 38]. A parallel tradition, exact geometric computation [41], decides discrete predicates exactly by filtering a floating-point evaluation through a proved error bound and falling back to exact arithmetic only when the filter fails [39, 40]. Both traditions give the same guarantee: a computed sign or bound is a proved statement about the real numbers, not an approximation of one.

The second foundation concerns what cannot be decided. The first-order theory of real closed fields is decidable [57], but adding the elementary functions makes it undecidable in general whether an expression in one real variable is nonnegative everywhere, or identically zero [58]. A sound procedure over expressions with exp, log or sin must therefore be allowed to return no answer, and the existing tools differ in how they treat that answer: δ-complete decision procedures weaken the question, returning "unsatisfiable" or "δ-satisfiable" for a perturbation δ chosen by the user [42, 43]; interval constraint propagation returns boxes of unknown status [46]. Three-valued logic gives the undetermined answer a semantics: in Kleene's strong logic [59] a conjunction is false if any conjunct is false, undetermined if none is false and some is undetermined, and true otherwise. The verdict lattice of Section 2.2.2 adopts this semantics, with refusal in the role of the undetermined value. The analogy is operational, not semantic: a refusal records that the procedure did not decide, not that the claim lacks a truth value.

The third foundation is the separation between finding an answer and checking it. A *certifying algorithm* returns, with each output, a witness

from which a simple and independent checker can confirm the output without trusting the algorithm [60]; proof-carrying code applies the same idea to programs [61], and SAT solvers ship proofs that small verified checkers replay [62, 63]. Sums-of-squares certificates found by semidefinite programming and checked in exact rationals [64, 65], and Taylor-interval certificates replayed inside a proof assistant [47, 48, 49], are instances in nonlinear real arithmetic. From statistical decision theory comes the last ingredient: a classifier that may abstain is evaluated by its risk on the cases it decides and by the fraction it decides, the error–reject tradeoff [66, 67, 68, 69]. Closest in intent is recent work on language-model outputs that builds test batteries sound by construction, so that a reported defect is never spurious, and states their detection limits [53]; cert-machine asks the same of every verdict, not only of refutations. The system combines these three lines: soundness from the inclusion property, a three-valued answer in which refusal is a result, and a witness for every verdict that an independent checker can re-run.

2.2. Mathematical formulation

2.2.1. Interval arithmetic and the inclusion property

Let $\mathbb{IR}$ denote the set of closed bounded real intervals $X = [\underline{x}, \overline{x}]$, with width $w(X) = \overline{x} - \underline{x}$ and midpoint $m(X) = (\underline{x} + \overline{x})/2$. A box is a vector $X = (X_1, \ldots, X_n) \in \mathbb{IR}^n$, with $w(X) = \max_i w(X_i)$. For $\circ \in \{+, -, \times, \div\}$ the interval operations are the set images $X \circ Y = \{x \circ y : x \in X,\ y \in Y\}$ ($0 \notin Y$ if $\circ = \div$), computed from the endpoints [17, 18]:

$$\begin{aligned} X + Y &= [\underline{x} + \underline{y},\ \overline{x} + \overline{y}], & X - Y &= [\underline{x} - \overline{y},\ \overline{x} - \underline{y}], \\ X \times Y &= [\min S,\ \max S],\ S = \{\underline{x}\underline{y}, \underline{x}\overline{y}, \overline{x}\underline{y}, \overline{x}\overline{y}\}, & X \div Y &= X \times [1/\overline{y},\ 1/\underline{y}]. \end{aligned} \tag{1}$$

Definition 1 (Interval extension)**.** Let $g : D \subseteq \mathbb{R}^n \to \mathbb{R}$. A map G from boxes in D to $\mathbb{IR}$ is an *interval extension* of g if $g(X) = \{g(x) : x \in X\} \subseteq G(X)$ for every box $X \subseteq D$ (the *inclusion property*), and *inclusion isotone* if $X \subseteq Y$ implies $G(X) \subseteq G(Y)$. This is weaker than Moore's extension, which also asks $G([x, x]) = g(x)$; it is what Neumaier [18] calls an inclusion function, and nothing below needs more.

The *natural extension* of an expression replaces each real operation by its interval operation and each elementary function by an enclosure of its range.

Theorem 1 (Fundamental theorem of interval arithmetic [17, 18])**.** *If every elementary function occurring in an expression for g has an inclusion-isotone interval extension, then the natural extension G of g is an inclusion-isotone interval extension of g.*

Theorem 1 is the only property that the soundness of a verdict requires (Proposition 2). Its price is overestimation: $G(X)$ may be much wider than $g(X)$, because each occurrence of a variable is treated as independent. Writing $q(A, B) = \max(|\underline{a} - \underline{b}|, |\overline{a} - \overline{b}|)$ for the Hausdorff distance between intervals, an extension has *convergence order* [70] ω on B if there are $\kappa > 0$ and $\eta \geq 0$ such that

$$q\big(g(X), G(X)\big) \leq \kappa\, w(X)^{\omega} + \eta \qquad \text{for every box } X \subseteq B. \tag{2}$$

In exact arithmetic $\eta = 0$. In floating point $\eta > 0$: outward rounding adds at least one ulp per operation and each function enclosure adds its tail bound, neither of which shrinks with $w(X)$, so η is the overestimation floor of the implementation on B. When every elementary function in the expression has a Lipschitz interval extension, the natural extension has order $\omega = 1$, and the mean-value form

$$G_{\mathrm{MV}}(X) = G\big([c, c]\big) + \nabla G(X)^{\top}(X - c), \qquad c \in X, \tag{3}$$

with c a double near $m(X)$, since $m(X)$ itself need not be representable, has order $\omega = 2$ [18], both with $\eta = 0$ in exact arithmetic. Higher-order enclosures, such as Taylor models [71], reduce the overestimation further. cert-machine uses the natural and mean-value forms, endpoint evaluation of monotone functions, the Krawczyk operator (Theorem 2) and second-order Taylor remainders in quadrature; it uses no Taylor models or affine forms. The reason is the claims it meets: they are low-dimensional, and either exact or decided with a margin that bisection reaches; the one family where the enclosure straddles the bar at the budget (the cosine minima of Section 3.2) is the place a higher-order form would help, and it is listed as future work rather than built. The order governs how much subdivision a decision costs (Proposition 3), never whether the decision is sound.

Outward rounding on doubles. IEEE-754 arithmetic in round-to-nearest returns, for each of $+, -, \times, \div$, the double nearest the exact result [72, 73, 74]. Evaluating Eq. (1) in round-to-nearest therefore does not enclose the exact endpoints, and the enclosure must be widened. JavaScript cannot set the rounding mode, so directed rounding is emulated by moving each computed endpoint one unit in the last place outward [75, 26].

Proposition 1. *Let $\circ \in \{+, -, \times, \div\}$, let A, B be intervals of doubles—pairs of non-NaN doubles with lower end at most upper end, an infinite endpoint allowed as a bound—with $0 \notin B$ when $\circ = \div$, and let y_-, y_+ be the round-to-nearest values of the smallest and largest endpoint combinations $a \circ b$. If no endpoint combination is indeterminate ($0 \cdot \infty$, $\infty - \infty$, ∞/∞), then $A \circ B \subseteq [\mathrm{down}(y_-), \mathrm{up}(y_+)]$, where down and up step one double toward $\mp\infty$ and fix $\mp\infty$; an exact result beyond the largest double rounds to $\pm\infty$, which the step fixes, so overflow is enclosed.*

The proof is in Appendix A. The domain of Proposition 1 is enforced, not assumed: a malformed operand or an indeterminate result is refused by exception and never decided; division by an interval containing zero is refused, not extended; a non-integer or overflowing integer exponent is refused, and a real power of a positive base is $\exp(p \log x)$. The trust base is stated once: the correctly rounded $+$, $-$, $\times$, $\div$ and square root that IEEE-754 mandates and the engine implements, its arbitrary-precision integers, and nothing in the platform's mathematical library, which ECMAScript leaves "implementation-approximated". Where the quantity is rational or a tie must be decided, exact arithmetic is used instead: reduced fractions of arbitrary-precision integers, to which doubles convert losslessly, with exact elimination that reports a singular matrix rather than a small pivot; where more precision is required, a separate arbitrary-precision arithmetic with true directed rounding, converted to doubles outward with an exact check.

Elementary and special functions. Theorem 1 requires an enclosure for every elementary function, and no library function is trusted as an enclosure, because ECMAScript gives no accuracy guarantee for the platform's exponential or logarithm—with one exception at the time of writing: an older helper that encloses the platform's cosine and sine by a derived pad is still called by six modules, among them the radii-polynomial certifier, and is scheduled for retirement (Section 4.5). Each function f is evaluated as a truncated series S_N in interval arithmetic plus an explicit bound R_N on the omitted tail, so that $f(x) \in S_N(x) + [-R_N, R_N]$, and the implementation must satisfy:

Assumption 1 (Function enclosures)**.** *For every function f of Table A.9 and every box X in its declared domain, either the call is refused or the returned interval $F(X)$ satisfies $f(X) \subseteq F(X)$, including near zeros of f.*

Appendix A states, for each function, the reduction, the series, the tail bound and the domain on which the bound holds (Lemmas 1 to 8), and how the tail is added: as an interval $[-R_N, R_N]$ (or $[0, R_N]$ for a one-sided tail) by an outward-rounded addition, the bound itself evaluated in interval arithmetic and its upper end taken, so that nothing is absorbed into the widening of Proposition 1. The kernel and its functions have been compared with Arb [27] on 1,012,121 seeded cases over 24 operations, with no enclosure that failed to contain the exact value (Section 3.2); that is evidence on sampled inputs, and the lemmas are the proof.

Long sums. A running interval sum rounds outward at each of n additions, so its width grows linearly in n; at the 175,320 hourly values of one buoy it is about a thousand times what the terms carry. Each endpoint stream $p_1, \ldots, p_n$ is therefore summed by the compensated algorithm Sum2 of Ogita,

Rump and Oishi [76, Proposition 4.5], whose result $\tilde{s}$ satisfies, also in the presence of underflow,

$$\Big|\tilde{s} - \sum_{i=1}^{n} p_i\Big| \le u\,\Big|\sum_{i=1}^{n} p_i\Big| + \gamma_{n-1}^{2} \sum_{i=1}^{n} |p_i|, \qquad \gamma_k = \frac{ku}{1-ku}, \quad u = 2^{-53}, \tag{4}$$

provided $nu < 1$ and no overflow occurs. The implementation evaluates the bound from computable quantities as $u|\tilde{s}| + 2\gamma_n^2 \sum |p_i|$, each step rounded upward, and adds it outward to each endpoint; the doubled term covers the difference between $u|\tilde{s}|$ and $u|\sum p_i|$ (supplement, S1), and both hypotheses are checked by name: a stream with $nu \ge 1$ or an overflowed partial sum is refused.

Derivatives. Gradients and Hessians come from second-order forward differentiation over the same interval arithmetic, or from univariate Taylor jets, so that the derivative a proof uses (e.g. ∇G in Eq. (3)) is an extension of the function the proof is about. Families with hand-written scores and Hessians evaluate them in the same arithmetic.

2.2.2. Claims, readings and verdicts

Boxes and claims. A box with rational endpoints is the object a claim is decided on; a double is a rational, so a box of doubles is such a box. A *declared envelope* B is a box each of whose sides carries the source of its bounds (a survey, a data sheet, a chart). A printed number is the box its digits allow: a decimal $\hat{d}$ printed with k fractional digits denotes

$$\big[\hat{d} - \tfrac{1}{2}10^{-k},\ \hat{d} + \tfrac{1}{2}10^{-k}\big], \tag{5}$$

so a printed 0.0634 is $[0.06335, 0.06345]$, and both endpoints are exact rationals. Equation (5) assumes that the printed value was rounded to nearest. The claim schema fixes that convention for a printed *value* and refuses any other box for it; a printed *bound* ($\le$, $<$, $\ge$, $>$) is taken as the exact rational its digits denote, and a value that was truncated or given to significant figures must be declared as an explicit rational box. Declared envelope ends are read as the nearest double, widened outward by one ulp unless the conversion is exact.

Definition 2 (Claim)**.** A *claim* is a pair (B, φ) of a domain B—a box, or a finite set of rational points, covered by its thin boxes—and a predicate

$$\varphi(x) \equiv \bigwedge_{j=1}^{m} g_j(x) \le c_j \tag{6}$$

whose functions have interval extensions G_j; an equality, or an exact predicate—the orientation of three points, or a Boolean combination of

exact integer tests such as the angular test of a kissing configuration—is the special case decided in exact arithmetic. It is read either *universally*, as $\forall x \in B : \varphi(x)$, or *epistemically*, as $\varphi(x^\dagger)$ for an unknown true input $x^\dagger \in B$.

The form is deliberately narrow: a conjunction of inequalities over one box, with disjunction allowed only among exact tests and no alternation of quantifiers. A statement outside it—an existential claim, an inequality quantified over a parameter that is itself unknown, a claim about an object that was not published—is not forced into the schema; it is refused with the reason *the instrument declined*, or marked *needs data* with the bytes that would decide it named (Table 1). Under the universal reading a single proved counterexample refutes the claim; under the epistemic reading it shows only that the declared evidence does not settle the question.

Definition 3 (Verdicts)**.** Let $\mathcal{X}$ be a finite cover of B by boxes. The claim is *certified* if $\sup G_j(X) \le c_j$ for every j and every $X \in \mathcal{X}$; *refuted*, under the universal reading, if some box $X^* \subseteq B$ has $\inf G_j(X^*) > c_j$ for some j, and under the epistemic reading if every $X \in \mathcal{X}$ has $\inf G_j(X) > c_j$ for some j; and *refused* otherwise. An epistemic refusal is *split* when two points $x^-, x^+ \in B$ have been proved to violate and to satisfy φ, and *unresolved* otherwise.

Proposition 2 (Soundness)**.** *If every G_j is an interval extension of g_j, then* CERTIFIED *implies $\varphi(x)$ for every $x \in B$;* REFUTED *implies $\neg\varphi(x)$ for some $x \in B$ under the universal reading and for every $x \in B$ under the epistemic reading; and a split* REFUSED *implies that neither $\forall x \in B\, \varphi(x)$ nor $\forall x \in B\, \neg\varphi(x)$ holds, so that $\varphi(x^\dagger)$ is not determined by the declared envelope.*

Proof. Let $x \in B$ and $X \in \mathcal{X}$ contain it. By the inclusion property $g_j(x) \in g_j(X) \subseteq G_j(X)$, hence $g_j(x) \le \sup G_j(X) \le c_j$ for every j. The refutations follow by the same argument with the inequality reversed, on X^* or on every member of the cover. For a split refusal, x^+ and x^- are points of B at which φ is proved true and false; either could be $x^\dagger$. □

In the engineering gates the box decided is the double hull $B^+ \supseteq B$ obtained by widening each declared end outward. Certification and epistemic refutation transfer from B^+ to B. A witness of a split refusal is a corner of B itself: the declared decimal, enclosed by its two neighbouring doubles and decided as that thin box, so that a verdict proved over the box is proved at the exact corner, which is a point of B. A witness that a floating-point search proposes (the rock-physics engine) is a double of B^+, within one ulp of B, and is reported as such.

Soundness says nothing about how often the procedure decides. The following result, in the spirit of δ-completeness [42], states when bisection is guaranteed to certify and what it costs.

Proposition 3 (Termination for robust claims)**.** *Let each G_j have convergence order ω with constants κ and η on B (Eq.* (2)*), and let the claim hold with margin $\varepsilon > \eta$: $g_j(x) \le c_j - \varepsilon$ for every j and every $x \in B$. Then every box $X \subseteq B$ with $w(X) \le ((\varepsilon - \eta)/\kappa)^{1/\omega}$ is certified, and bisection of the widest coordinate certifies the claim at depth at most*

$$d^* = n\left\lceil \log_2\Big(w(B)\,\big(\kappa/(\varepsilon-\eta)\big)^{1/\omega}\Big)\right\rceil^+, \tag{7}$$

with at most 2^{d^*} boxes in the final cover.

Proof. By Eq. (2), $\sup G_j(X) \le \sup g_j(X) + \kappa w(X)^\omega + \eta \le c_j - \varepsilon + \eta + \kappa w(X)^\omega \le c_j$ when $\kappa w(X)^\omega \le \varepsilon - \eta$. Under widest-coordinate bisection, every n consecutive splits along a branch at least halve $w(X)$: a coordinate wider than half the current width is split before any narrower one, and at most n such coordinates exist; after d^* splits $w(X) \le 2^{-d^*/n} w(B) \le ((\varepsilon - \eta)/\kappa)^{1/\omega}$. □

Proposition 3 explains the refusals the evaluation must expect. A claim whose margin does not exceed the floor ($\varepsilon \le \eta$; in particular $\varepsilon = 0$, a bound attained at a point) is never certified by bisection alone and needs one of the certifiers of Section 2.2.4 or a higher precision that lowers η; and the box count grows as $(w(B)(\kappa/(\varepsilon-\eta))^{1/\omega})^n$, so a depth budget $d_{\max}$ turns high-dimensional or nearly tight claims into refusals by budget rather than into wrong verdicts. The depth bound is proved for widest-coordinate bisection, the rule of the universal branch-and-bound. The engineering gates choose the coordinate by sensitivity instead (the largest floating-point variation of the model across the box; or, after the one-signed coordinates are fixed at the deciding end, the largest $|\partial g/\partial x_i(X)|\, w(X_i)$, the monotonicity test and a smear-type rule [29, 77]), and for those rules no depth bound is claimed. The rule changes cost and termination, never soundness.

Composition. Order the verdicts as REFUTED $\prec$ REFUSED $\prec$ CERTIFIED. A record with several checks $\varphi = \bigwedge_i \varphi_i$ receives

$$v(\varphi) = \min_{\prec}\{v(\varphi_i)\}, \tag{8}$$

that is, REFUTED if any check is refuted, otherwise REFUSED if any is refused, otherwise CERTIFIED—the strong conjunction of Kleene's three-valued logic [59].

Corollary 1. *Under Proposition 2, if all checks φ_i are decided on the same box B and under the same reading, the composition rule Eq.* (8) *is sound for φ under that reading.*

Proof. If every φ_i is certified, $\varphi_i(x)$ holds on B for every i, hence φ. Under the universal reading a refuted φ_i gives $x \in B$ with $\neg\varphi_i(x)$, hence $\neg\varphi(x)$; under the epistemic reading it gives $\neg\varphi_i(x)$ for every $x \in B$, hence $\neg\varphi$ on all of B. □

A split refusal does not survive conjunction: if φ_1 holds only on one half of B and φ_2 only on the other, each is split while $\varphi_1 \wedge \varphi_2$ is false on all of B. A composed refusal is therefore unresolved unless its witnesses are proved for φ itself. In the implementation the composition is computed from the checks' words alone and carries no witness, so a composed refusal is never labelled split; the split refusals of the record are issued by the two engineering gates at the level of one check, a single inequality, and a receipt that composes several checks carries that check's witnesses beside the composed word, never as its own. Every check of a record shares one box and one reading; items of one record that are decided on different objects are not composed by Eq. (8) but reported as MIXED with the counts of each verdict (Table 1); a record with refuted and refused items composes to REFUTED by Eq. (8) and carries both counts.

A refusal is terminal: it is never retried at lower rigour, never converted into a probability and never dropped from the record. The only path from REFUSED to CERTIFIED is a new exact argument—a finer cover, an extension of higher order, another certifier, or the bytes the claimant had not yet published. The three verdicts are not probabilities, and there is no fourth.

Witnesses. Every verdict carries the object that justifies it, in the sense of a certifying algorithm [60]. A certification carries its *certificate*: the inputs pinned by digest, the hull of its enclosures with the counts of its cover, and a statement of what would make it fail (the *falsifier*); the constructor refuses a certificate without a stated falsifier, a proved verdict without evidence, and a refusal without a reason. A refutation carries the falsifying object, printed; a split refusal its two witnesses, decided as thin boxes by the same enclosures. An exact certificate re-checks without the system's code; an interval certificate records the hull and the counts of its cover, not the cover, and is re-checked by re-deciding the claim from the pinned inputs (exporting the cover is future work).

The public register composes the three verdicts into words that say what they are made of (Table 1); none is a fourth verdict. The register's builder refuses a row that carries no scope sentence or no kind, so a CERTIFIED is never published without its scope; that the kinds of refusal (instrument declined, claimant published nothing, budget exhausted, box did not close, audit reach stated) are never summed into one total is a rule of reporting, not yet a check in code.

2.2.3. Flip thresholds

A split refusal leaves open a question that has a size. Let the claim depend on a declared scalar p—an operating choice, or one end of an interval that a measurement could narrow—through boxes $B(p)$ and predicates φ_p, and write $\mathsf{T}(p)$ for "$\varphi_p(x)$ holds for every $x \in B(p)$" and $\mathsf{F}(p)$ for "$\varphi_p(x)$ fails for every $x \in B(p)$".

Table 1: Verdicts and their compositions in the public register [14].

word	composed of	witness carried
CERTIFIED	—	certificate with its falsifier; an exact one re-checks without the engine
REFUTED	—	falsifying object, printed
REFUSED	—	reason; for a split refusal, two witnesses and the thresholds
PARTIAL	certified + refused	which fragment certified and which core was not reached
NEEDS DATA	refused, cause with claimant	the bytes that would decide the row
MIXED	certified + refuted, one record	counts of both and the items refuted
REPAIRED	refuted + certified	object as printed refuted; object beside it certified
QUEUED	not a verdict	nothing; not counted as decided

Assumption 2 (Monotone truth)**.** $\mathsf{T}(p)$ *implies* $\mathsf{T}(p')$ *for every* $p' \le p$*, and* $\mathsf{F}(p)$ *implies* $\mathsf{F}(p')$ *for every* $p' \ge p$*. A sufficient condition is that the boxes do not depend on* p*,* $B(p) = B$*, and every* $g_j(x; p)$ *is nondecreasing in* p*.*

Proposition 4. *Let Assumption 2 and the hypothesis of Proposition 2 hold, let* $B(p)$ *be nonempty, and define* $p_T = \sup\{p : \mathsf{T}(p)\}$ *and* $p_F = \inf\{p : \mathsf{F}(p)\}$*, so that* $p_T \le p_F$*. If the procedure, at a value* m *of the parameter, certifies the claim, then* $\mathsf{T}(p)$ *holds for every* $p \le m$ *and* $p_T \ge m$*; if it refutes it epistemically, then* $\mathsf{F}(p)$ *holds for every* $p \ge m$ *and* $p_F \le m$*; if it returns a split refusal, then* $p_T \le m \le p_F$*.*

The proof, the closed form where p enters additively ($g(x; p) = p + h(x)$ on a fixed box: $\mathsf{T}(p)$ for every $p \le c - \overline{h}$ and $\mathsf{F}(p)$ for every $p > c - \underline{h}$, from one enclosure $[\underline{h}, \overline{h}]$ of h over B, every step in directed rounding), the second case—a measurement that narrows one side of the box, where the half of Assumption 2 for T fails and the nesting of Proposition 7 replaces it—and the way each running instance verifies the hypotheses are in the supplementary material (S2). The threshold turns a refusal into a measurement or an operating change with a known effect on the verdict: at this pump pressure the rule is proved, and with the roughness shown to be at least this value it is proved too [1].

2.2.4. Certifiers

Each certifier turns a question that is analytic in form into finitely many decisions of the kind Proposition 2 covers (Table 2), and most exist precisely for the claims that Proposition 3 cannot reach: zeros, maxima and bounds attained with no margin. None is new; the contribution is that each is mapped to the grammar with an explicit witness and explicit refusal reasons.

Appendix B states the result each rests on and the supplementary material (S3) its implementation; here is the catalogue.

Table 2: Certifiers, the result each rests on, the witness it publishes and the reasons for which it refuses. For a formal proof, the statement that the proof proves is read against the theorem claimed, not decided.

question	result used	witness	refusal reasons
a finite set, exhaustively	every element decided; a Cauchy–Schwarz bound, asserted on the data before any path runs, makes the floating-point products exact	counts; any violating element	set beyond the budget; bound not met
an entropy certificate	exact pushforwards; each $-p \ln p$ enclosed	enclosure of the ratio	enclosure straddles the bar; malformed law
a zero, exactly one	Krawczyk operator (Theorem 2); radii polynomial for fixed points	$K(X) \subseteq \operatorname{int} X$; or $p(r) < 0$ and contraction factor < 1	no contraction within the round limit or below the radius cap
a maximum in a box	strict concavity (Proposition 5)	Hessian minors or nonsingularity proof	no contraction; Hessian not proved negative definite; climb stopped at the family's boundary
a supremum at a limit	mean value theorem (Proposition 6)	neighbourhood, score sign	sign not proved
every input in a box	bisection (Proposition 3); monotone implicit roots; multilinear corners (Lemma 9)	cover; corner witnesses	budget exhausted (split or unresolved)
a sign, exactly	static filter, exact fallback [39, 40]	sign, or ON	literal beyond fixed precision
a polynomial ≥ 0	rational SOS identity [64, 65]	q_i, $c_i \geq 0$, zero residual	negative weight; nonzero residual (printed)
min of a cosine polynomial	Chebyshev form, Sturm chain, interval Newton, Markov bound	enclosure of the minimum; algebraic certificate	straddles the bar; isolation or refinement fails; algebraic: minimum not interior
a formal proof, built	kernel acceptance at the pinned toolchain, replayed by a second kernel	build log; axioms reported	resources (stack, memory, time)

Proposers. Every certifier takes its candidate from a floating-point search—Nelder–Mead and Levenberg–Marquardt for a likelihood, Newton for a root, sampling for a minimum, a semidefinite solver for a decomposition, a language model for a construction—and none of these is in the trust path. A proposer may be wrong, slow or adversarial; provided Assumption 1 holds and the certifier implements its theorem correctly, it can cost a refusal, never

a false verdict.

2.3. Comparison metrics and decision rules

2.3.1. Metrics

Verdict–truth confusion matrix. Each benchmark claim $i = 1, \ldots, N$ carries a ground-truth label $\tau_i \in \{\text{true}, \text{false}, \text{malformed}\}$—it holds on all of B, it fails at a proved witness, or it is not a well-formed instance of its claim schema—established by an oracle that is independent of cert-machine and itself exact (Section 2.3.3). It receives a verdict $v_i \in \{\textsc{certified}, \textsc{refuted}, \textsc{refused}\}$, a refusal being split or unresolved. The confusion matrix $N_{v\tau} = \#\{i : v_i = v,\ \tau_i = \tau\}$ is the primary result. Proposition 2 predicts structural zeros in the cells that would be errors:

$$N_{\textsc{certified},\text{false}} = 0, \quad N_{\textsc{refuted},\text{true}} = 0, \quad N_{\textsc{certified},\text{malf.}} = N_{\textsc{refuted},\text{malf.}} = 0, \tag{9}$$

the last pair because a malformed object must be refused by its shape check and never decided. Any nonzero entry in Eq. (9) is a defect in code or in a hypothesis, and is reported individually. Under the epistemic reading a fourth label, *undetermined* (the predicate holds at one proved point of B and fails at another), has the split refusal as its correct answer; the benchmark of Section 3 contains universal and exact claims only, so that label does not arise there.

Upper bound on the error rate. Observing zero errors on N_{F} false claims does not show that the error probability θ is zero. The one-sided Clopper–Pearson bound [78] at level $1 - \beta$ after x errors is the beta quantile $\theta_U = \text{Beta}^{-1}(1-\beta;\ x+1,\ N_{\text{F}}-x)$, which at $x = 0$ is $1-\beta^{1/N_{\text{F}}} \approx 3/N_{\text{F}}$ for $\beta = 0.05$, the "rule of three" [79]. The same bound is reported for false refutations on true claims. The bound presumes independent draws from a population of claims, whereas the benchmark is assembled from chosen records and injected faults, and a defect in one code path affects every claim of its class; it is therefore read as a description of this benchmark, not as an error rate that transfers. Proposition 2 is what the system claims mathematically; the bound measures the gap between the code and the theorem on the claims tested.

Coverage and selective risk. A sound three-valued decider is a selective classifier with a reject option [66, 67, 68], in which a split refusal is an informative answer and only an unresolved refusal is an abstention. Writing $D = N - N_{\textsc{refused}_{\text{u}}}$. for the number of informative answers, its *coverage* is $C = D/N$ and its *selective risk* R is the number of errors of Eq. (9) divided by D. Because the mix of labels is set by the benchmark's design, C is also reported per label: the certification rate on true claims, the refutation rate on false claims and the refusal rate on malformed ones, each by claim class,

with the unresolved refusals broken down by reason (never summed across reasons). Soundness predicts $R = 0$; the measured R tests the implementation against that prediction, and the performance of a sound implementation is then carried by C. A floating-point checker is placed on the same risk–coverage plane [67, 69]: it has $C = 1$ and $R \geq 0$; cert-machine is predicted to have $C \leq 1$ and $R = 0$.

Expected cost of a decision. With a cost c_e for acting on a wrong verdict, c_s for the measurement a split refusal asks for and c_r for the argument an unresolved refusal asks for, the expected cost per claim is $L = c_e\, R\, C + c_s\, S + c_r\, (1 - C)$, S the fraction of split refusals, in the spirit of Chow's error–reject tradeoff [66]: $c_s S + c_r(1 - C)$ for cert-machine against $c_e R_{\mathrm{fp}}$ for a floating-point checker, both measured in the benchmark, so the break-even ratio c_e/c_r is reported as a function of the proportion of false claims—which the injected faults inflate—rather than assumed.

Tightness, cost and agreement. For a certified enclosure F the number of certified decimal digits is $\delta(F) = -\log_{10}(w(F)/|m(F)|)$ for $0 \notin F$; for a split refusal the thresholds of Section 2.2.3 say how far an operating change or a measurement must go. For each decision the harness records wall-clock time, CPU time and peak memory, cold and warm, under two profiles (a portable one with a wall-clock cap, an extended one without), and the certificates keep the boxes evaluated and the depth reached. Verdicts are deterministic, so agreement with a second implementation is exact agreement, field by field or by digest, and a single disagreement is a defect; against an established library the primary result is the table of containment per operation. A paired comparison with another decider (McNemar's test [80] on the discordant counts) is defined for the day one can express the same claims, and none does.

2.3.2. Decision rules

Table 3 lists every choice the system makes when it decides a claim, with the rule and the result that justifies it. The first two rows are modelling choices, declared with the claim: they fix *what* is decided, and a different choice decides a different claim. Every other rule either is forced by a result above or can change only the cost of a decision, never its soundness.

2.3.3. Evaluation design

The evaluation answers Q3 with a benchmark of known truth and answers Q1 and Q2 with observational records.

Benchmark with known ground truth. The benchmark is the system's own public register, mutated. Each register row that carries a published finite object—a tensor decomposition, a point configuration, a set of circles or points, a printed constant, a polynomial map, an entropy law, a truth

Table 3: Decision rules and their mathematical basis.

decision	rule	basis
reading a printed number	a value: the box of half a unit in the last printed place; a bound: the exact rational of its digits	Eq. (5)
universal or epistemic	universal if the claim quantifies over inputs; epistemic if it concerns one unknown true input	Definition 2
arithmetic	exact rationals if the object is rational or a tie must be decided; outward-rounded intervals otherwise	Proposition 1, Theorem 1
extension	natural form (engineering gate); natural and mean-value forms Eq. (3) intersected on every box (rock-physics engine)	Eq. (2); cost only
split or stop	widest coordinate (universal branch-and-bound); largest sensitivity (engineering gates); stop at a depth of 12 splits per branch (gate), a depth and a box budget (branch-and-bound, by default 9 and 2×10^6), or 6,000 evaluations (rock physics)	Proposition 3; cost only
which certifier	by the question's form (Table 2)	the theorem of each row
composing checks	minimum in REFUTED $\prec$ REFUSED $\prec$ CERTIFIED	Eq. (8), Corollary 1
kind of refusal	split if both witnesses are proved; unresolved otherwise	Definition 3
threshold	closed form when the parameter enters additively; otherwise one-sided bisection over nested boxes (sixteen steps in the gate, ten in the rock-physics engine)	Proposition 4, Proposition 7
accepting a battery	all red controls fire	protocol rule R2 (Section 2.4.2)

table—is the *original*; from it the benchmark builds *mutants* by transformations of three kinds, each with a label known by construction: *false* (a coefficient raised by one, a digit moved, a point displaced, a bar raised one unit past the certificate's own printed end), *true* (an invariance of the claim: a translation, a rotation, a relabelling of coordinates), and *malformed* (a missing exponent, a weight off a probability law, a digit beyond the alphabet). Every label is established by an *oracle* written for the benchmark in the Python standard library—exact fractions, directed decimal rounding, integer convolutions—that shares no code with the deciders it judges, and a false mutant counts only when the oracle exhibits the violating element (a *witnessed* false claim); a mutant the oracle cannot witness is discarded and counted as such. The design, the classes, the kill switch (the campaign

stops if fewer than half the rows yield a witnessed false mutant) and the outcome words were pre-registered before the first mutant ran, the note is pinned by digest in every record, and the one amendment, dated, moved no definition. This is the planted-solution design of benchmarks for other verifiers [81] applied to the system's own record. Every claim is also decided by a baseline: (B0) a floating-point checker that evaluates the predicate in double precision at a tolerance $\tau \in \{10^{-6}, 10^{-9}, 10^{-12}\}$, as producers' checkers typically do. The metrics of Section 2.3.1 are reported for cert-machine and for B0.

The arithmetic against an independent library. The interval kernel's operations and functions are compared, case by case on seeded inputs, with the exact value or range computed in Arb [27]: the rule is containment, and two planted wrong kernels must be caught. This answers Q2 on sampled inputs; the lemmas of Appendix A answer it on paper.

Coverage across domains, and the register. Q1 asks whether one grammar covers claims from mathematics, engineering and metocean statistics. It is answered by reporting, for each domain, how many claims could be stated in the form of Definition 2 and with which verdicts they were decided, and by listing the claims that could not be so stated. The register of machine-generated claims [14] is reported as an observational study: how many claims of each source certify, refute or refuse, and what kinds of defect the refutations expose. It measures what independence finds in practice, not a false-certification rate, since the ground truth of its claims is not known independently; its three largest exercises are the subjects of the companion papers.

2.4. Software architecture

2.4.1. System overview

The mathematics of Section 2.2 is implemented in four layers (Fig. 1): an arithmetic layer (exact rationals, outward-rounded intervals, function enclosures, compensated sums and derivatives; Section 2.2.1); a certifier layer (the certifiers of Table 2 and the decision loop); a record layer (certificates, witnesses, digests and the register's builder); and a verification layer (the batteries, red controls and second implementations of Section 2.4.2). The repository holds 86 instrument directories, about 201 thousand lines of JavaScript and 126 thousand of Python, lifted code included. A *proposer*—a search, a model, a language model or a printed table—is never trusted: its output is read as a claim over a box and decided by a *certifier*. A floating-point *screen* may discard candidates before certification but can never admit one. The verdict, its witness and the inputs pinned by SHA-256 form a *record*: an exact record re-checks without the engine, and an

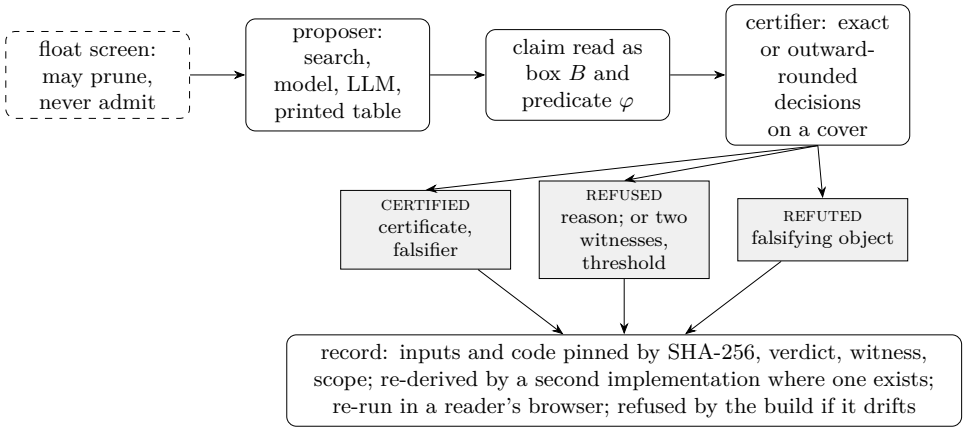


Figure 1: The cert-machine pipeline. The proposer is never opened and never trusted; its output is decided by a certifier, and the verdict, its witness and the pinned inputs form a record; an exact record re-checks without the engine, an interval record by re-deciding from its pinned inputs.

interval record is re-checked by re-deciding the claim from its pinned inputs (Section 2.2.2).

The decision loop that Definition 3 describes—decide the corners of B as thin boxes, build the cover by bisection, return the verdict with its witness—is sketched in the supplementary material (S4). It is the loop as implemented in the engineering gate (epistemic reading) and in the universal branch-and-bound, which share no code; the corner step is the gate's, and the branch-and-bound has none.

2.4.2. Verification protocol

Proposition 2 makes the verdicts as sound as the code that computes the extensions. The protocol below is how that code is guarded. Each rule is an executable gate—a test battery that exits non-zero, a page builder that will not write its page, a numbers generator that will not write a paper's macros—and Table 4 states which gates are enforced at every level and which are not yet.

R1. Screens prune, never admit.. A floating-point screen may discard candidates; only the certifier can record a hit, and the campaign runner plants known answers that must pass every stage before a run may start [82, the methods note].

R2. Adversarial controls must fire.. A *red control* is a deliberately wrong variant of a test that must be rejected; a battery whose red controls do not all fire fails. One example: a "gate" evaluating the model only at the centre of a box, which the corner rule must catch. Red controls are a form of mutation testing [83, 84, 85], by which soundness defects have been found

in SMT solvers [86, 87]; of the 164 batteries the test target runs (103 in JavaScript, 61 in Python), 99 state their red controls, 705 in all, and the others check without one.

R3. Pin the bytes.. Every claim is decided against a byte sequence named by path and SHA-256 and re-hashed at decision time; certifying modules are pinned in a ledger; a page refuses to build when a pinned input changed, when the record was written by another version of its verifier, or when its battery fails.

R4. Independence from the claimant.. The claimant's program is never in the trust path; the decision is re-derived from the published statement and bytes. The precise form adopted is that the system *shares no code with the claimant*, and every exact certificate re-checks with no engine present; the exceptions known at the time of writing are listed in Table 4.

R5. Second implementations.. Where a verdict is the product, a program written from the specification in another language and arithmetic, sharing no code, re-derives it: the benchmark's oracles, the kissing samples, the entropy and truth-table certificates, the step functions, the trace-constant and Zhang–Zagier certificates, δ_3, three engineering decisions (supplement, S5). Sharing no code does not make failures independent: independently written programs fail together more often than independence predicts [88], and the same has been measured for programs generated by language models [89, 90], which drafted parts of cert-machine as well as many of the claimants' artefacts; agreement is evidence against defects the two do not share, not proof of their absence.

R6. Records re-run.. Published records re-run in the reader's browser by the bytes that wrote them and are compared with the record by digest or field by field. A rerun kit of 31 commands re-derives the register's records; run whole on 2026-10-21, it moved none of the 132 rows, and the pre-registered self-audit of Section 3 ran 124 runs of its 31 commands under two profiles and found every re-derived record identical to its pin outside declared time stamps (23 records and 125 rows without a wall-clock cap; 18 records and 113 rows under a 60-second cap, the other 5 refused for resources).

R7. Honest counting.. Every number in a paper is a macro written from a record by a generator that refuses when the record no longer supports the sentence quoting it; queued rows are not decided rows; refusal kinds are never merged.

2.4.3. Reproduction

Every constant in this paper is interpolated from a record in the cert-machine repository at commit aae61990d737, so that the document cannot quote a number the machine did not certify; it will

Table 4: Enforcement status of the protocol rules at the time of writing.

rule	enforced per battery / page / record	enforced in aggregate
R1 screens	yes	partly: every family of the generation loop decides through one verdict module; instruments and applications keep their own verdict words, checked against the closed list at publication
R2 red controls	yes	yes: the test target exits non-zero on any failing battery and the control page refuses to build (an explicit override prints the red rows)
R3 pinned bytes	yes	yes
R4 independence	by design	no: two instruments reuse the system's own code across a producer–checker line (the mean-field-game capacity certifier imports a frozen earlier verifier; the lemniscate certifier shares the interval kernel with the bench it was lifted from)
R5 second implementations	where a second program exists (supplement, S5)	not for every certifier (point-in-polygon, cosine minimum, sums of squares)
R6 re-run	yes	re-runs the record, not an independent code path
R7 counting	yes	yes

be archived with the next release of the repository, under the concept doi:10.5281/zenodo.22225860 (the previous version is in release v2026.10.21, doi:10.5281/zenodo.23551746). The headline results re-derive with the commands of the supplementary material's reproduction table (S5), and the references are checked too: every bibliography entry is resolved at its source by the build's reference check, and the build refuses an entry with no verified row.

3. Results

3.1. The benchmark with known ground truth

Design, as pre-registered. The register held 132 rows from 25 records when the pre-registration was pinned on 2026-10-21. Of these, 119 rows belong to a family whose decider the benchmark can call on a modified object; 97 of them carry a published finite object and were mutated, and 22 were not, each with its reason recorded in the campaign's record: rows decided by reading a finite object that the claimant did not publish in a mutable form (the counterexample rows), and rows whose object is a bound certified from

a literature argument rather than from bytes (the Turán, HorizonMath, GNNW and sum–product rows, and one kissing row with no object). The mutation classes of Section 2.3.3 produced 572 mutants: 322 false, 153 true and 97 malformed. The oracle witnessed 295 of the false ones; the other 27, all perturbations of printed coordinates by 10^{-12} in the point-set family, moved the object below the precision its claim is printed at, so the oracle could not exhibit a violation and the benchmark discards them, as pre-registered. The kill switch was not tripped: 94 of the 119 rows yielded a witnessed false mutant, against a bar of 60.

Table 5: The benchmark by family (`certs/selfaudit-mutations.json`): rows of the register in the family, rows with a mutable object, mutants by label (false/true/malformed), and the outcomes that would be errors or abstentions. "Not as required" counts the mutants whose decision word differs from the pre-registered required word without being an error: a false claim answered REPAIRED (the printed bound refuted and the certificate's own value certified beside it), one answered UNWITNESSED, and the one malformed term the partial path of a polynomial-map decider read silently (Section 4.5).

family	rows	objects	mutants	F/T/M	false cert.	false ref.	refused (F)	discarded	not as req.
matrix-multiplication tensors	45	45	315	180/90/45	0	0	3	0	0
kissing configurations	11	10	69	39/20/10	0	0	0	0	0
point sets and shapes (EinsteinArena)	28	28	136	80/28/28	0	0	3	27	18
a printed constant (Erdős 852)	1	1	5	2/2/1	0	0	0	0	0
polynomial maps and fields	9	9	35	17/9/9	0	0	0	0	1
counterexamples (finite objects by reading)	14	0	0	0/0/0	0	0	0	0	0
entropy certificates and a truth table	4	4	12	4/4/4	0	0	0	0	0
Turán-type bounds	2	0	0	0/0/0	0	0	0	0	0
HorizonMath rows	3	0	0	0/0/0	0	0	0	0	0
the GNNW constant	1	0	0	0/0/0	0	0	0	0	0
a sum–product bound	1	0	0	0/0/0	0	0	0	0	0
all	119	97	572	322/153/97	0	0	6	27	17+1+1

The verdict–truth matrix. Table 5 is the result. Of the 295 witnessed false claims, 271 were refuted, 6 were refused (a Gaussian coefficient bound too large for exact double summation in three tensor rows; three flat-polynomial rows whose coefficients, perturbed off the integers, lie outside the decider's schema), 17 were answered REPAIRED—the register's composition of a refutation of the printed bound with a certification of the certificate's own value, which is a refutation of the claim as printed and not an error—and 1 was answered UNWITNESSED, a refusal under the ledger's own word. Of the 153 true mutants, 153 were certified, and so were all 97 originals. Of the 97 malformed objects, 96 were refused and 1 was decided: the structural zeros of Eq. (9) hold for false certifications and false refutations and fail once for a malformed object, and that one case is reported below as an open finding. The selective risk is $R = 0$ on the decided claims; the coverage on witnessed false claims is 271 refuted and 17 repaired of 295, and the unresolved refusals are 6 by resources and 1 by the decider's own word. With zero false certifi-

cations on 295 witnessed false claims the one-sided Clopper–Pearson bound at $\beta = 0.05$ is $\theta_U = 1.01\%$, and with zero false refutations on 250 true claims (mutants and originals) it is 1.19%; both describe this benchmark and its classes, not other claim sets (Section 2.3.1).

Two defects found by the benchmark's rule. The campaign's first run, at commit d961cda2 on 2026-10-21, did not show this matrix. It showed 28 false certifications and 49 malformed objects decided, and both were defects of the system. The first: the ledger of point-set configurations composed its word from whether the published bytes witnessed a value, while the agreement of the printed digits with that value was a field beside the word, gated only by the page build; a printed value moved one unit in its last digit past the bytes therefore stayed CERTIFIED on every row with a printed value. The fix is one module that composes the word in one place—a witnessed object whose printed value the exact value does not reproduce under the row's convention is MISPRINTED, the register's REFUTED with the kind *not from the published data*—and the whole ledger was re-derived under it with no row changing word. The second: the tensor and kissing deciders answered a malformed object (a tensor with the wrong number of slices, a configuration that is not n vectors in $\mathbb{R}^d$, a weight row without its twenty weights) with the verdict word REFUTED, the identity having failed; the fix makes every shape check answer REFUSED with its reason, and the one identity that fails under both tensor layouts stays REFUTED. Both fixes are pinned in the campaign's record with the commit each was found at and every mutant it covers, the campaign was re-run whole after them, and Table 5 is that re-run. The pre-registration's rule that a defect found by the campaign is fixed on the record and the campaign re-run, never the definition moved, was followed; the one amendment to the note is dated and lists what it does not change.

The conventional check, on the same claims. The baseline B0 evaluates each predicate in double precision at a tolerance. At $\tau = 10^{-6}$ it accepts 27.8% of the witnessed false claims (82 of 295), at 10^{-9} 11.2% (33) and at 10^{-12} 2.4% (7). The tolerance cuts both ways: at the same three settings it confirms only 94, 81 and 72 of the 97 true originals, and 149, 136 and 127 of the 153 true mutants, because four of the nine polynomial maps are decided by no tolerance at all (their determinants are exact integers that double precision does not hold). On the risk–coverage plane of Section 2.3.1 the checker has $C = 1$ and R between 2.4% and 27.8% of the false claims at the tolerances producers use; the decider has $R = 0$ and C as above.

Concordance, and the hosted rows. Where a register row is decided by two or more programs—the deciders under test, the benchmark's oracles, the rerun kit's standard-library verifiers, the second implementations of rule R5—the concordance record compares their answers field by field: 91 comparisons on 56 rows, 39 at the level of the enclosure or the violating count and 49 at

the level of the word, with no disagreement. The 416 kernel replays of the mathematics release ran on hosted runners and are carried in as recorded, never merged into the laptop's table: 398 CERTIFIED under two kernels, 18 REFUSED for resources, wall-clock from 18 to 12,453 seconds (median 587) and peak memory up to 13.4 GiB.

Cost. The mutation campaign took 17 minutes on the laptop (Apple M2, 8 cores, 16 GB, Node 24.14.1); the re-derivation of the register's records under the extended profile 26 minutes of wall-clock and under the portable profile 9, with a peak resident memory of 0.66 GiB, every run cold and warm with its wall, CPU and memory in the record. Nothing was rented.

What the benchmark does not cover. Every claim in it is universal or exact, so the epistemic reading, the split refusal and the undetermined label are not measured; the engineering gate's planted faults (7 faulty proposals, 6 refuted and 1 refused [1]) are the only ground truth of that kind and are targeted, not sampled. The mutation classes were chosen by the author, so the benchmark bounds the defects it can find by the classes it plants; a defect outside them is invisible to it. No second decider that can express these claims exists, so the paired comparison of Section 2.3.1 is not reported.

3.2. The kernel against an independent library

The interval kernel was run on 1,012,121 seeded cases across 24 operations—up to 50,000 each, drawn from full-precision doubles, subnormals, arguments near overflow and underflow, and boxes that straddle a function's critical points—and every enclosure it returned was compared with the exact result, or for a box the exact range, computed in Arb [27] at 256 bits and more (Table 6). Every one of the 1,009,497 enclosures it returned contained the exact value, and it refused the other 2,624; none of the 12 malformed inputs (a NaN, an inverted interval, division through zero, a non-integer or overflowing power, a logarithm at or below zero) was decided; and the comparison caught two planted wrong kernels, a round-to-nearest addition in 19,846 of 50,000 cases and the platform's exponential at the ends in 50,000 of 50,000. The audit covers the special functions the likelihood certifier uses, the gamma function by Spouge's formula, the Bessel function, the inverses of the normal and incomplete-gamma distributions—each judged against a bracket bisected in Arb on certain signs—and the compensated sum against the exact rational sum of its terms. The widths tell where the enclosures are loose: the gamma function's tenfold Spouge margin and the Bessel series near the zeros of J_ν are contained by many ulps, which costs tightness, never soundness.

What the method refuses that a stronger method could decide. Incompleteness has a size, and it is reported. Of the 416 Comparator challenges of the mathematics release, 18 are refused for a stack, memory or time limit

operation	cases	contained	violations	refused	median ulps
$x + y$	50,000	50,000	0	0	2
$x - y$	50,000	50,000	0	0	2
$x \times y$	50,000	50,000	0	0	2
x/y	48,529	48,529	0	0	2
x^2	50,000	50,000	0	0	2
$\lvert x \rvert$	50,000	50,000	0	0	0
x^k	50,000	50,000	0	0	9
exp	48,289	46,799	0	1,490	120
log	49,838	49,838	0	0	5
sin	50,000	50,000	0	0	365
cos	50,000	50,000	0	0	328
tanh	50,000	50,000	0	0	0.5
the platform's cos, padded	50,000	50,000	0	0	4,816
the platform's sin, padded	50,000	50,000	0	0	4,820
$\ln \Gamma$	50,000	50,000	0	0	46
ψ	50,000	50,000	0	0	36
ψ'	50,000	50,000	0	0	22
$P(a, z)$	50,000	50,000	0	0	7,061
Φ	49,465	49,465	0	0	117
Γ (Spouge)	50,000	48,866	0	1,134	21,638,632
J_ν	5,000	5,000	0	0	1,098,797
Φ^{-1}	5,000	5,000	0	0	240
P^{-1}	5,000	5,000	0	0	5,934
Sum2, n terms	1,000	1,000	0	0	—

Table 6: The interval kernel against Arb (`certs/kernel-arb-audit.json`): for each operation, the cases run, the enclosures that contain the exact value computed in Arb, the violations, the in-domain refusals, and the median width of a point input's enclosure in units in the last place of the larger end of the result (the record holds the maxima too; a median below one ulp, as for tanh, comes from results just below 1; no median for Sum2, whose inputs are streams). The widths of the padded platform functions are those of the older helper that Section 4.5 names.

that more resources might clear. Of the 1,650 cosine-minimum decisions of the generation loop, 71 are refused because the enclosure straddles the bar, which a finer cover or a higher-order form might separate. The kernel's 2,624 refusals near underflow and overflow are a design choice: the enclosure $[0, 2^{-1074}]$ would be valid at underflow, and the kernel does not return it. None of these is converted by lowering the rigour; each stays refused with the reason that stopped it, until a stronger method decides it.

3.3. The register, and its three largest exercises

The public register holds 132 claims from 59 claimants, decided from 25 records between 2026-08-25 and 2026-10-05 and frozen on 2026-10-27; decisions made after the freeze (the registry replays, the kissing configurations of the second exercise) are not register rows. Of its rows 131 are decided and 1 is queued, not counted: 107 CERTIFIED, 9 PARTIAL, 2 REFUTED, 5 MIXED, 6 REPAIRED and 2 NEEDS DATA. 24 rows carry a kind of defect other than

none; the most frequent is *narrower scope* (8 rows), a formal statement or a check that covers less than the claim beside it. Of the 107 rows whose claimant is a machine or a machine-assisted pipeline, 17 carry a defect; of the other 24, 7. The register measures what independence finds, not an error rate: no row's truth is known independently of the decision. Three exercises dominate it and are the subjects of the companion papers.

A configuration of points. An autonomous discovery system published new lower bounds for the kissing number in 19 dimensions (25, 27, 32–39, 43, 45 and 49–55), each a configuration of integer or algebraic vectors, their sizes proved by moment identities and the angular condition by structure, with the release's own files recording that the pairwise inner products were not checked [10]. Two configurations were published as exact objects and decided at once; the other seventeen were NEEDS DATA with the missing files named, and the claimant published them within days [91]. Every pair was then decided exactly: 9.75×10^{15} pairs, 6.72×10^{12} at exactly sixty degrees and none closer, in three arithmetics each exact on its data—single-precision BLAS on the laptop, exact algebraic integers for the two closed-form constructions, and half-precision tensor cores on one rented GPU for the seven largest (6.8 GPU-hours, $8.15 billed), the floating-point paths made exact by a Cauchy–Schwarz bound asserted on the data before any path runs. All 19 bounds hold; the claimant linked the check from its release. The companion paper gives the materials, the calibration, the second implementation and the limit of the GPU argument [82, the kissing companion].

A curated record. An optimization-constants registry marks with an asterisk a bound whose verification is minimal and states what removes the mark [92, 93]. Three asterisked bounds rest on finite certificates—two entropy laws for the sums–differences constants and a truth table for the Fourier entropy–influence constant—and were decided twice from the registry's bytes by programs that share no code with each other or with the claimants; all three are CERTIFIED with their ratios enclosed. The registry merged the replay and removed the marks, and its pages cite the replay [82, the registry companion]. The registry's open queue offered 10 further bounds: 7 are CERTIFIED by a computation made here, one on an earlier two-kernel record, one was checked by reading alone and is counted as a reading, and 1 is REFUSED—a package that certifies its own assembly and not the analysis it assembles, with the three inputs that would decide it named. One bound was printed on the wrong side of its own certificate, $C_{1d} \le 1.463071582607559$ against the certificate's exact $1.46307158260755932746\ldots$; the record holds the printed bound REFUTED and the repaired one CERTIFIED. The certificate's own checker had printed OK on all 4 of the claims it was given, of which 2 are refuted here.

A formal release. A mathematics release of 719 machine-written manuscripts came with 416 Lean challenge statements as its formal spine, and with a Navier–Stokes repository proved against a statement written months earlier [12, 11]. Every challenge was replayed under Lean's kernel and under a second kernel written independently of it: 398 accepted by both, 18 REFUSED for resources, none rejected. The Navier–Stokes development built clean at its pinned commit with the standard axioms, and its formal statement, read hypothesis by hypothesis against the paper's theorem, lacks the energy clause of that theorem; the record is PARTIAL at 8937a8f4 with the clause named in its scope, and the next upstream commit, f9e8bc5b of 2026-09-10, proves it. Of the 242 manuscript families whose statements were read against their theorems, 127 match and 34 state less than, or other than, what they claim; the readings were made by AI agents and are not verdicts; their recall was measured once, on 12 defects planted in short statements, 12 flagged. The companion paper gives the kernel lane, the readings and the 14 gaps the audit pinned [82, the release companion]; the Navier–Stokes reading has its own note.

3.4. Coverage across domains

Q1 asks whether one grammar covers claims from mathematics, from engineering and from metocean statistics. Table 7 lists, for each domain, the claims stated in the form of Definition 2 and their verdicts, each read from the record that holds it. In mathematics the register's rows are the claims. In engineering the flow-line gate decides nine demonstrator receipts over a declared box of six inputs under the epistemic reading—the lead proposal split, with the pump pressure at which the rule is proved (204.0 bar) and refuted (222.3 bar) and the roughness measurement that would prove it (27.000 mm) reported as thresholds—and seven faulty proposals planted beside it, six refuted and one refused [1]; the rock-physics engine decides detectability over declared petro-elastic boxes under the same reading [55]. In metocean statistics the likelihood certifier decided 46,602 fits of six families at 2,589 open-sea cells of a public hindcast: 40,778 CERTIFIED (87.5%), 5,824 REFUSED, of which 5,751 are the generalized gamma proved at its log-normal limit with the proof; at the benchmark's buoys 353 of 402 fits are CERTIFIED and every ranking of families is decided or refused by name [2]. What could not be stated in the grammar is listed with the register: a claim about an object the claimant did not publish is NEEDS DATA, and a statement decided by reading alone—a formal statement against its theorem, an inventory of a package—is recorded as a reading and never as a verdict.

Table 7: Claims decided by one grammar, by domain, each count read from the record that holds it. The engineering and metocean rows are the companions' headline counts; the register's composite words are grouped.

domain, instrument	claims	CERTIFIED	REFUTED	other	other, by word
mathematics, the register	132 rows, 59 claimants	107	2	9+5+6+2+1	partial, mixed, repaired, needs data, queued
engineering, the flow-line gate	7 planted faults; one lead proposal	0	6	1+1	refused (split, with thresholds)
metocean, the likelihood certifier	46,602 atlas fits	40,778	0	5,824	refused, 5,751 at the lognormal limit with the proof
metocean, the benchmark's buoys	402 fits	353	0	402 less 353	refused by name

4. Discussion

4.1. What the results show

Soundness, measured. Proposition 2 predicts zero false verdicts, and on 572 claims of known truth the implementation produced zero, after two defects that the benchmark's own rule exposed were fixed on the record. Neither defect was in the arithmetic. One was in a word: a ledger that said CERTIFIED about the bytes while the printed digits beside them disagreed. The other was a verdict word given to an object that was not a claim at all. Both are the verifier's kind of defect—a gap between what was checked and what the word said—and both were found by planting claims against the system's own register, which is what the protocol asks of claimants. The lemmas of Appendix A, written for this version, found three more of the same kind in the arithmetic (Section 4.5): a thin box of a double standing for π, a tail bound short by its own prefactor, a margin computed in the platform's floating point; each lay below the enclosures' widths and each was unproved until written down.

The defects in the record are the verifier's kind. No bound of the three exercises was found false and no theorem was false. One registry bound is printed on the wrong side of its own certificate; a pairwise condition was proved by structure and never checked pair by pair; a certificate's checker compares in double precision what it encloses at a hundred digits; a package certifies its own assembly and not the analysis it assembles; a clause of a theorem is absent from the formal statement, and formal statements are narrower than the theorems beside them. Each was closed, or named, by a finite computation or a recorded reading whose only hard part was deciding to make it.

Refusal is a result, and it has a size. Seventeen kissing rows were NEEDS DATA from the request until the points were released, each naming the files that would decide it; the request was answered and the rows closed, an

opacity measured and then removed. The gate's split refusal carries the pump pressure at which it would become a proof. The package is REFUSED with its three missing inputs named. The Navier–Stokes reading issues no verdict and defers to the kernel, so the missing clause became a scope sentence rather than an accusation. Every refusal says what would convert it, and the kinds are never summed. The cost of exactness, by contrast, is small: the objects are integers, rationals and bounds, and the one expensive decision—10^{15} pairs—spent its budget on the size of the set, not on rigour.

A decision changes the record. Each exercise ends outside the system. The kissing claimant published its points in the form the check needed and linked the decision from its release. The registry's maintainer, who had written down beforehand what removes the mark of an unverified bound, removed it on the replays and cites them. The release's next commit, made the day after the audit pinned the repository, proves the clause the reading found missing. We draw no causal conclusion from these cases. They show what the grammar makes possible: a CERTIFIED with its certificate can be cited, a REFUSED that names its files can be answered, and a scope sentence can be closed.

4.2. Independence, counted

"Independent" carries different weights, and they are not interchangeable. Table 8 separates five levels of evidence and places every result of this paper on them. Every computed result reaches the first level; most reach the second; three reach the third, each for one part of a claim only: the kernels' claim, the interval kernel's arithmetic and the entropy deciders' arithmetic; none reaches the fourth or the fifth. Outside parties have since re-derived 7 records of the repository (3 parties, 2026-09-30 to 2026-10-24, each with the same verdicts; `corpus/external-reruns.json`): 5 of them on rows of this paper's exercises — the registry's C_{3b}, C_{3c}, C_{42}, C_{84b} and C_{31} rows and the kissing ledger, by 2 parties with their own code or with the tools that write the records — and 2 outside it, the $\lambda(4)$ audit [94] and the Ramsey graphs. Those on the exercises are third-level evidence for the parts re-derived, obtained after the audits were posted; one of them read the C_{42} decider and found three defects in what it printed, none moving a verdict, fixed and recorded in the benchmark's amendments. The registry's merge is not on the table at all: its maintainer applied his rule to the replay and did not re-derive it, so it is an outcome, not a check.

4.3. Relation to existing practice

Validated numerics and computer-assisted proof are the certifiers' ancestry (Section 2.1). Formal proof checks a proof against a stated theorem [95, 96, 97], independent type checkers replay the kernel's work [98, 99], and a translation that compiles may still not be the theorem intended [15, 51];

level	evidence	where it holds in this paper	what it cannot catch
1	the same code, rerun from pinned inputs	every record re-read against its pins at build; the register re-derived by the rerun kit on 2026-10-21 (0 rows moved) and by the self-audit under two profiles (23 records identical outside stamps)	any defect in the code itself
2	a second implementation by the same author, from the same specification	the benchmark's six oracles, the kissing samples, the entropy and truth-table certificates, the step functions, the trace-constant and Zhang–Zagier certificates, δ_3, the Hankel cross-checks, three engineering decisions; written by AI agents from one brief	a misreading the two share
3	an implementation by another party	the Lean kernel and a second kernel for the kernels' claim (Section 3.3); Arb for the interval kernel's arithmetic on 24 operations and for the entropy deciders' arithmetic, on sampled inputs (Section 3.2); 5 outside reruns on the exercises' rows (C_{3b}, C_{3c}, C_{42}, C_{84b}, C_{31}, the kissing ledger), same verdicts; outside this paper, the $\lambda(4)$ audit and the Ramsey graphs	a defect outside the part re-derived
4	review of the specification and the scope by a specialist	none: the readings were made by AI agents and no specialist has confirmed them	—
5	a formal proof of the metatheory	none: Proposition 2 and the lemmas are paper proofs; the exact-class certificates are in a form a proof assistant could check	—

Table 8: Five levels of evidence, and where each result of this paper stands on them. They are counted apart and never summed.

autoformalization is now a research area of its own [100, 101]. Certifying algorithms return a certificate that a small independent checker verifies [60], as SAT solvers do with proofs that verified checkers replay [62, 63], and proof-carrying code ships a proof of a safety policy with the program [61]. Decision procedures already treat a failure to decide as an answer [102, 103, 43, 104], and reproducibility initiatives re-run published computations [105, 106]. Benchmarks for machine mathematics grade answers against keys [107]; the system's companion on grading measures what tolerance rules accept [82, the certified-evaluation companion]. Each of these enforces one of the properties the system combines—a recorded refusal, a certificate that a small checker verifies, a reproducible build, independence from the producer—and

none is organised around other parties' published claims; the combination, enforced by code that refuses, is what is claimed.

4.4. Threats to validity

- *Selection.* Every claim was chosen by the author: the register records every row as self-initiated, and the exercises were picked because they had finite objects to decide. The results describe these claims, not machine-generated mathematics at large.

- *The benchmark's classes.* The mutation classes were designed by the author, so the benchmark can find only the defects its classes reach; the Clopper–Pearson bounds describe this benchmark and do not transfer. Every claim in it is universal or exact, so the epistemic grammar is measured only by the gate's targeted faults.

- *The author in the outcomes.* The outcomes of Section 3.3—the points published, the marks removed—followed requests and pull requests the author made. They are reported as what happened, and no causal claim rests on them.

- *Correlated implementations.* The deciders, their second implementations and the benchmark's oracles were written by AI agents under the author's direction, from shared briefs. N-version programs written that way can fail together [88, 89, 90], and level 2 of Table 8 is counted apart for that reason.

- *The arithmetic's evidence.* The comparison with Arb covers sampled inputs, not every input; the lemmas cover the functions, not the code that implements them; no IEEE 1788 conformance suite [21] has been run on the interval layer.

- *The readings.* The statement readings and the Navier–Stokes reading were made by AI agents and checked only by second readings made the same way; their recall was measured once, on short statements under an alert brief, and not on the long ones. No human specialist has confirmed them.

- *The GPU argument.* The exactness of the seven largest kissing decisions rests on the documented single-precision accumulation of tensor cores [108, 109, 110], characterised for earlier devices than the one rented, on the bound asserted before the run, and on a calibration that reproduced the laptop's exact contact counts on two configurations; the calibration is a check, not a proof.

4.5. What is not claimed, and the limits

Not claimed. Interval arithmetic, the Krawczyk operator, floating-point filters, sums of squares and Sturm sequences are not claimed; they are cited. The validity of a declared model is not claimed: a PROVED engineering number holds for the declared model over the declared envelope, not for the sea or the reservoir. A kissing bound "decided here" is a statement about one published configuration, not about the record in that dimension. Nothing is certified in the regulatory sense; the word is used in the sense of Definition 3 only. Nothing is a probability.

Locality and incompleteness. A certified maximum is the maximum in its box; a global statement over a family would need a branch-and-bound over the family's whole parameter space, which the system does not run. The system refuses what it cannot decide, and the refusal rate is a property of each campaign, published with it (Section 3.2).

Readings are not decisions. Three results of the register rest in part on readings rather than computations: the clause found absent from the Navier–Stokes formal conclusions, the statements of the mathematics release found short of their claims, and the inventory of the record package. Each reading was produced by AI agents from a written brief, and no human specialist has confirmed them. Until one does, the gaps are reported as what the readings found, with their recall as measured.

Soundness rests on code. Proposition 2 is a theorem about interval extensions; whether a program computes one is a fact about the program. The defences—second implementations, red controls, pinned bytes, calibrations, the planted-claim benchmark, the comparison with Arb, outside reruns—reduce that risk and do not remove it. What a read of the code found on 2026-10-09 was closed the same day with the correction published beside the old numbers (the generation loop's closed-form hunt had counted 54.6 million float comparisons as refutations; every refutation is now a proved disjointness). What the benchmark found on 2026-10-21 is in Section 3.1. What writing the lemmas of Appendix A found the same day was closed before this version was built, and every record that pins the changed modules was re-derived: the square root of π in the error function was taken from a thin box of the double nearest π, which does not contain π; the error function's tail was padded by the bare next term, short of the alternating-series bound by its own factor $2/\sqrt{\pi}$; the gamma function's error margin was a platform float; the four transcendental tails were evaluated in round-to-nearest and covered only by the margin below the sum's ulp; the lower end of $\ln\Gamma$ across its minimum was a hard-coded double; the flip thresholds were rounded to nearest before the decimal rounding; the gate's witnesses were the widened doubles, one ulp outside the declared box; and the kissing

runner recorded the exactness bound it did not assert. Each lay below the widths in play, and none moved a published verdict; each was a place where the code said less than the paper. What remains open, by the system's own rule that a check not made is not a check:

- the older helper that encloses the platform's cosine by a derived pad, contained in every case of Table 6 but wide, is still called from six files, among them the radii-polynomial certifier, whose records are re-run when it is retired;
- the partial path of one polynomial-map decider reads a term with a missing exponent silently and answers PARTIAL where its full path refuses (the one malformed object decided in Table 5);
- one quadrature routine, rigorous on dyadic cells of $[0, 1]$, is called on another interval by one instrument; one battery labels some positive assertions as red controls; in the engineering demonstrator a PROVED consistency check means that the claim meets an outer enclosure;
- two of the system's own instruments reuse its code across a producer–checker line (Table 4), and the same-author second implementations are not independent re-derivations (Table 8);
- only the exact-class certificates are in a form a program other than this system's can check; an interval certificate records the hull and the counts of its cover, not the cover.

5. Conclusions and recommendations

cert-machine decides the numbers and claims that machines propose with three verdicts, each carrying the object that justifies it; the interval and exact verdicts are sound under the inclusion property, which the arithmetic is built to secure and whose function enclosures now carry a stated lemma each; its certifiers reduce the questions that arrive—a zero, a maximum, a statement for every input, a sign, a nonnegative polynomial, a minimum, a formal proof—to finitely many exact or outward-rounded decisions; its protocol keeps them honest by code that refuses; and its record is public and re-runs. Measured against 572 claims of known truth, it produced no false verdict and refused 6; measured against an independent library on 1,012,121 cases, every enclosure held; pointed at 132 machine-generated claims, it found the verifier's kind of defect, and the claimants, the registry and the release moved on what it found. Two of its own defects were found by the same rule and are on the record. The three questions are answered as far as the record allows: one grammar covers the three domains (Q1); the hypothesis holds on the inputs tested and is proved on paper for the functions used (Q2); and on claims whose truth is known the system certifies what is true,

refutes what is false and refuses a stated fraction by resources, at a cost of minutes on a laptop (Q3).

Recommendations. To producers of machine-generated mathematics: publish the object, not only its description—the points of a configuration in one integer form, the witness a theorem rests on, the analytic inputs a certificate freezes, the data a fit was made from—with a digest beside each file; state the scope of every check in the release itself; and when a formal statement is narrower than the theorem it accompanies, say so where the theorem is stated. To those who maintain the tables and records of a field: distinguish a bound that has been decided from outside from one that has only been claimed, and write down what removes the distinction. To those who build checkers: plant claims of known truth against your own record, with an oracle that shares no code with the decider, before anyone else does.

Declaration of competing interest

The authors declare no competing financial or personal interests. The claimants named in Section 3.3, the maintainer and contributors of the optimization-constants registry, and the maintainers of the benchmarks and tables named here have not been consulted on this draft; the exchanges with the kissing-number group and with the registry took place on the public threads cited.

CRediT authorship contribution statement

Carlos Toledo: conceptualization, methodology, software, validation, investigation, data curation, writing—original draft, writing—review and editing. Pedro Veras Guimarães: methodology, formal analysis, writing—review and editing. The roles are to be confirmed by the co-authors.

Data availability

Every record, certifier, figure source and table source is public in the cert-machine repository (`https://github.com/carlostoledo1891/cert-machine`, commit aae61990d737; archived as [82]): the register `certs/claims-ledger.json`; the benchmark's pre-registration `notes/selfaudit-preregistration-2026-10-20.md`, its four records `certs/selfaudit-*.json` and its oracles under `instruments/selfaudit/`; the kernel audit `certs/kernel-arb-audit.json`; the records of the exercises named in the companions. The claimants' bytes are cited at their sources and pinned by SHA-256 digest in the records; they remain under their owners' licences. The code is MIT-licensed.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work the author used large language models (Claude, Anthropic) to draft code, exposition and this manuscript in a certified-verification setting: constructions and prose were proposed with the help of the models, and every quantitative claim is decided by outward-rounded interval arithmetic or exact rational arithmetic, with adversarial controls that must fire for a record to be accepted. After using the tool the author reviewed and edited the content and takes full responsibility for the content of the publication. The records, not the models, are the evidence offered; they are public and re-runnable. For this paper specifically: the deciders, the certifiers' code, the benchmark's oracles and the readings named in Section 3.3 were produced by AI agents (Claude) under written briefs and consolidated into the public records; no verdict here rests on a reading alone, and no human specialist has confirmed the readings (Section 4.5). A referee-style reading of the previous version by an AI agent, and the co-author's restructure of the methodology with its line-by-line code checks, were applied in this version.

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Appendix A. The arithmetic in detail

Proof of Proposition 1. For these operations the extremes of $a \circ b$ over $A \times B$ are attained at endpoint combinations (monotonicity in each argument on each sign region, with $0 \notin B$ for division). The exact extreme lies within half the spacing of the doubles on its side of the rounded value, and the adjacent double on that side is a whole spacing away—also when the rounded value is a power of two, where the spacing below is half the spacing above, and in the subnormal range—so it lies on the correct side of $\mathrm{down}(y_-)$ and $\mathrm{up}(y_+)$. □

Table A.9 lists every elementary and special function the certifiers call, with the lemma that establishes Assumption 1 for it; the lemmas are stated here and proved in the supplementary material (S1). Throughout, a series is evaluated by the natural extension on the enclosed argument, the tail bound is evaluated in interval arithmetic and its upper end $\overline{R}$ taken, and $[-\overline{R}, \overline{R}]$ (or $[0, \overline{R}]$) is added by an outward-rounded addition, so the bound is never absorbed into the widening of Proposition 1. The enclosures of π and $\ln 2$ are exact rationals (Machin's formula; $\mathrm{atanh}(1/3)$) converted outward, and every constant below ($\sqrt{\pi}$, $\sqrt{2\pi}$, $\pi/2$) is derived from them.

Table A.9: Elementary and special functions, how each enclosure is obtained, the lemma that proves it, and whether the function is among the 24 operations compared with Arb (Section 3.2; evidence on sampled inputs, not a proof).

function	method	proof	Arb
exp	reduction $x = k\ln 2 + r$, $\lvert r\rvert < 1$; 24 Taylor terms; geometric tail	Lemma 1	yes
log	$x = \mathrm{m}\,2^e$; series of atanh in $v = (\mathrm{m}-1)/(\mathrm{m}+1) \in [0, 1/3)$, one-sided tail	Lemma 2	yes
$\sin, \cos$	reduction by an enclosure of $\pi/2$; 16 Taylor terms, Lagrange tail; a box's range forced to ± 1 where an extremum may lie	Lemma 3	yes
$\ln 2, \pi$	exact rationals via atanh(1/3) and Machin's formula, converted outward	exact	through exp, log
Γ	Spouge's formula, $a = 20$, $z > 0.4$; the published bound, tenfold, enclosed; wide boxes by unimodality	Lemma 4	yes
$\ln\Gamma, \psi, \psi'$	the argument lifted above 20 by the recurrences; Binet's series to $m = 8$; the first omitted term as bound; boxes by monotonicity, across the minimum by convexity	Lemma 5	yes
$P(a, z)$	series with a geometric tail; the prefactor in logarithms; boxes by monotonicity in both arguments	Lemma 6	yes
J_ν	power series by exact term ratios, 40 terms, a certified ratio bound	Lemma 7	yes
erf, erfc, Φ	erf by its alternating series for $\lvert z\rvert \le 2$, the next term times $2/\sqrt{\pi}$ as bound; erfc for $x \ge 1$ by Laplace's continued fraction, consecutive convergents bracketing the value	Lemma 8	yes
Φ^{-1}, P^{-1}	bisection that moves an end only on a certified sign	monotonicity	yes
Sum2	the compensated sum with the bound of Eq. (4) added outward	Ogita et al. [76, Prop. 4.5]	yes

Lemma 1 (exp)**.** *Let x be a double, k an integer, and $R \ni x - k\ln 2$ the interval computed from x, k and the enclosure of $\ln 2$ by Proposition 1, with $\rho = \max_{r\in R}\lvert r\rvert < 1$ and $-1074 \le k \le 1023$ (the call is refused otherwise). Then for every $r \in R$, $e^r \in \sum_{i=0}^{24} r^i/i! + [-T, T]$ with $T = \rho^{25}(25!)^{-1}(1-\rho/26)^{-1}$, and $e^x = 2^k e^{x-k\ln 2}$ exactly, 2^k being a double.*

Lemma 2 (log)**.** *Let $x > 0$ be a normal double, $x = \mathrm{m}\,2^e$ with $\mathrm{m} \in [1, 2)$ (exact), and $V \ni v = (\mathrm{m}-1)/(\mathrm{m}+1)$ the interval computed outward, with $\nu = \max V < 0.34$ (refused otherwise). Then $\ln x = e\ln 2 + 2\operatorname{atanh} v$ and $\operatorname{atanh} v \in \sum_{k=0}^{39} v^{2k+1}/(2k+1) + [0, T]$ with $T = \nu^{81}/(81(1-\nu^2))$.*

Lemma 3 (sin, cos)**.** *Let x be a double, H the enclosure of $\pi/2$, n an integer and $R \ni x - n\pi/2$ the interval computed outward from x, n and*

H, with $\rho = \max_{r\in R}|r| \le 2$. Then $\sin x$ and $\cos x$ are $\pm\sin r$ or $\pm\cos r$ according to $n \bmod 4$, and for $|r| \le 2$, $\sin r \in \sum_{j=0}^{15}(-1)^j r^{2j+1}/(2j+1)! + [-\rho^{33}/33!, \rho^{33}/33!]$ and $\cos r \in \sum_{j=0}^{15}(-1)^j r^{2j}/(2j)! + [-\rho^{34}/34!, \rho^{34}/34!]$. For a box X of width below 2π, $\sin(X)$ lies in the hull of the images of its ends, extended to 1 (to -1) whenever some $\pi/2 + 2k\pi$ (some $3\pi/2 + 2k\pi$) may lie in X, the test made on the enclosures of these points and answering "may" whenever it cannot separate; a box of width at least 2π is enclosed by $[-1, 1]$; likewise for cos *with the critical points $2k\pi$ and $\pi + 2k\pi$.*

Lemma 4 (Γ by Spouge's formula)**.** *Let $a = 20$ and $w \ge 0$ real. With $c_0 = \sqrt{2\pi}$ and $c_k = (-1)^{k-1}(a-k)^{k-1/2}e^{a-k}/(k-1)!$ for $1 \le k \le a-1$, $\Gamma(w+1) = (w+a)^{w+1/2}e^{-(w+a)}\big[c_0 + \sum_{k=1}^{a-1} c_k/(w+k) + \varepsilon_a(w)\big]$ with $|\varepsilon_a(w)| \le a^{-1/2}(2\pi)^{-(a+1/2)}$ [111]. The implementation encloses each c_k, the sum and the prefactor by the natural extension and widens the bracket by the upper end of $10\,a^{-1/2}(2\pi)^{-(a+1/2)}$ computed in interval arithmetic, a tenfold margin; for $0.4 < z < 1$ it uses $\Gamma(z) = \Gamma(z+1)/z$, and for a box of width above 10^{-3} the log-convexity of Γ with its minimum in $[1.4616, 1.4617]$.*

Lemma 5 ($\ln\Gamma$, ψ, ψ')**.** *For real $z \ge 20$ and $m = 8$, with B_{2k} the Bernoulli numbers,*

$$\ln\Gamma(z) = (z-\tfrac12)\ln z - z + \tfrac12\ln 2\pi + \sum_{k=1}^{m}\frac{B_{2k}}{2k(2k-1)z^{2k-1}} + R_0,$$

$$\psi(z) = \ln z - \frac{1}{2z} - \sum_{k=1}^{m}\frac{B_{2k}}{2k\,z^{2k}} + R_1, \qquad \psi'(z) = \frac{1}{z} + \frac{1}{2z^2} + \sum_{k=1}^{m}\frac{B_{2k}}{z^{2k+1}} + R_2,$$

with $|R_0| \le |B_{2m+2}|/((2m+2)(2m+1)z^{2m+1})$, $|R_1| \le |B_{2m+2}|/((2m+2)z^{2m+2})$ and $|R_2| \le |B_{2m+2}|/z^{2m+3}$. An argument below 20 is lifted by the recurrences in interval arithmetic. Over a box, ψ is increasing and ψ' decreasing; $\ln\Gamma$ is convex with its minimum at $x^* = 1.4616\ldots$, and across x^* the lower end is $\ln\Gamma(p) - |\psi(p)|\max(p - \underline{x}, \overline{x} - p)$ for any $p > 0$, from the enclosures at p.

Lemma 6 (the regularized incomplete gamma)**.** *For $a > 0$ and $z > 0$, $P(a,z) = z^a e^{-z}\Gamma(a+1)^{-1}\sum_{n\ge0} t_n$ with $t_0 = 1$ and $t_{n+1} = t_n\, z/(a+n+1)$; if $r_N = z/(a+N+1) < 1$ then $0 \le \sum_{n>N} t_n \le t_N\, r_N/(1-r_N)$. Over a box P increases in z and decreases in a; $P^{-1}(a,p)$ increases in both.*

Lemma 7 (Bessel J_ν)**.** *For $\nu > -0.6$ and $x > 0$, $J_\nu(x) = \sum_{m\ge0} t_m$ with $t_0 = (x/2)^\nu/\Gamma(\nu+1)$ and $t_{m+1} = -t_m(x/2)^2/((m+1)(\nu+m+1))$; if $q = (x/2)^2/((M+2)(\nu+M+2)) < 1$ then $|\sum_{m>M} t_m| \le |t_{M+1}|/(1-q)$. For J'_ν, whose terms carry the factor $(2m+\nu)/2$, the ratios beyond $M = 40$ are at most $1.026q$, and for $q \le 0.9$ the tail is at most $2|t'_{M+1}|/(1-q)$. A box $[0,x]$ with $x \le 1$ is enclosed by $[\max(0, 1 - x^2/4), 1]$ for $\nu = 0$ and by $[0, (x/2)^\nu/\Gamma(\nu+1)]$ for $\nu > 0$.*

Lemma 8 (erf, erfc)**.** *For $|z| \le 2$ and $N = 40$,*

$$\operatorname{erf} z = \frac{2}{\sqrt{\pi}}\Big[\sum_{k=0}^{N} \frac{(-1)^k z^{2k+1}}{k!\,(2k+1)} + T\Big], \qquad |T| \le \frac{|z|^{2N+3}}{(N+1)!\,(2N+3)},$$

because the terms decrease in magnitude from $k = 2$ on. For $x \ge 1$, $\sqrt{\pi}\, e^{x^2} \operatorname{erfc} x = 1/(x + \frac{1}{2}/(x + 1/(x + \frac{3}{2}/(x + \cdots))))$, Laplace's continued fraction [112, 7.9.2], whose consecutive convergents bracket its value [113]; the implementation takes the hull of the 80th and 81st.

Appendix B. The certifiers' results

Theorem 2 (Krawczyk [23, 18, 24])**.** *Let $f : D \subseteq \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable, $X \subseteq D$ a box, $\hat{x} \in X$, Y a nonsingular matrix and $F'(X)$ an interval extension of the Jacobian over X. If $K(X) = \hat{x} - Y f(\hat{x}) + (I - YF'(X))(X - \hat{x}) \subseteq \operatorname{int} X$, then f has exactly one zero in X.*

Proposition 5. *Let ℓ be twice continuously differentiable on a convex box X, let $\nabla \ell$ have exactly one zero θ^* in X, and let the Hessian of ℓ be negative definite at every point of X. Then θ^* is the unique maximiser of ℓ on X.*

Proposition 6. *Let $\ell(\mu, \sigma, Q)$ be continuous on $B \times [0, q_1]$ and differentiable in Q on $B \times (0, q_1]$, with $\partial \ell / \partial Q < 0$ there. Then $\ell(\mu, \sigma, Q) < \ell(\mu, \sigma, 0)$ for every $(\mu, \sigma) \in B$ and $Q \in (0, q_1]$.*

Lemma 9. *Let f be multilinear on a box $X \subset \mathbb{R}^n$. Then $f(X) = [\min_v f(v), \max_v f(v)]$, the minimum and maximum taken over the 2^n vertices v of X.*

Proposition 7. *Let the hypothesis of Proposition 2 hold and let the boxes be nested, $B(p') \subseteq B(p)$ for $p \le p'$. If the claim is certified over $B(p_0)$, then $\mathsf{T}(p)$ holds for every $p \ge p_0$ and $\inf\{p : \mathsf{T}(p)\} \le p_0$; the same holds for an epistemic refutation and F.*

How each is implemented—the Krawczyk rounds and radius cap, Sylvester's criterion and the nonsingularity proof, the series in the shape parameter at the lognormal limit, the monotone implicit functions, the filtered orientation predicate with its error bound [114, Thm. 2.2 and §3.1], the exact sum-of-squares identity, the Chebyshev–Sturm minimum, the two claims of a formal proof, the threshold procedures and the decision loop—is in the supplementary material (S2–S4), with the proofs of the propositions above and the reproduction table (S5).

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